EXAMPLE 7 *Draw a rough sketch of the curve* $y = \cos^2 x$ *in* $[0, \pi]$ *and find the area enclosed by the curve, the lines* $x = 0$, $x = \pi$ *and the x-axis.*

SOLUTION We prepare a table for the values of $\cos^2 x$ at different points between 0 and π as given below.

x	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$	$2\pi/3$	$3\pi/4$	$5\pi/6$	π
$\cos^2 x$	1	0.75	0.5	0.25	0	0.25	0.5	0.75	1

By plotting these points and joining them with a free hand, we obtain a rough sketch of $y = \cos^2 x$ as shown in Fig. 20.11. The required region is the shaded region in Fig. 20.11.

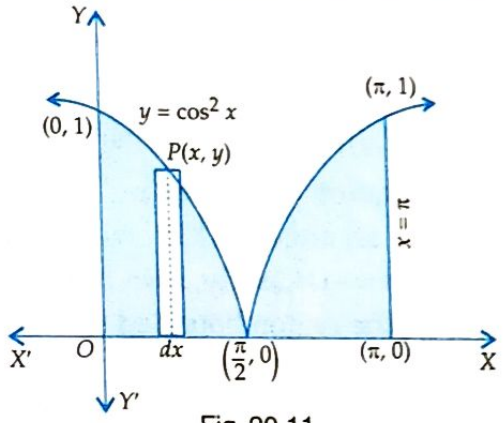


Fig. 20.11

To find the shaded region, we slice it into vertical strips. Each vertical strip has its lower end on x-axis and the upper end on $y = \cos^2 x$ So, the approximating rectangle shown in Fig. 20.11 has, length $= |y|$, width $= dx$ and area $= |y|\, dx$. Since the approximating rectangle can move from $x = 0$ to $x = \pi$. So, required area A is given by

$$A = \int_0^{\pi} |y|\, dx = \int_0^{\pi} y\, dx \qquad [\because\ y \geq 0\ \therefore |y| = y]$$

$$\Rightarrow \quad A = \int_0^{\pi} \cos^2 x\, dx \qquad [\because\ P(x, y) \text{ lies on } y = \cos^2 x]$$

$$\Rightarrow \quad A = \frac{1}{2}\int_0^{\pi} (1 + \cos 2x)\, dx = \frac{1}{2}\left[x + \frac{\sin 2x}{2}\right]_0^{\pi} = \frac{1}{2}\left\{\left(\pi + \frac{\sin 2\pi}{2}\right) - 0\right\} = \frac{\pi}{2} \text{ sq. units}$$

EXAMPLE 8 *Find the area of the region bounded by the line* $y = 3x + 2$, *the x-axis and the ordinates* $x = -1$ *and* $x = 1$. [NCERT]

SOLUTION Let $P(x, y)$ be any point on the line $y = 3x + 2$. The approximating rectangle shown in Fig. 20.12, has length $|y|$ and width dx. Clearly, it can move between $(-1, 0)$ and $(1, 0)$. So, required area A is given by

$$A = \int_{-1}^{1} |y|\, dx = \int_{-1}^{-2/3} |y|\, dx + \int_{-2/3}^{-1} |y|\, dx$$

$$\Rightarrow\ A = \int_{-1}^{-2/3} -y\, dx + \int_{-2/3}^{1} y\, dx \qquad \left[\because y < 0 \text{ for } -1 < x < -\frac{2}{3} \text{ and } y > 0 \text{ for } -\frac{2}{3} < x < 1\right]$$

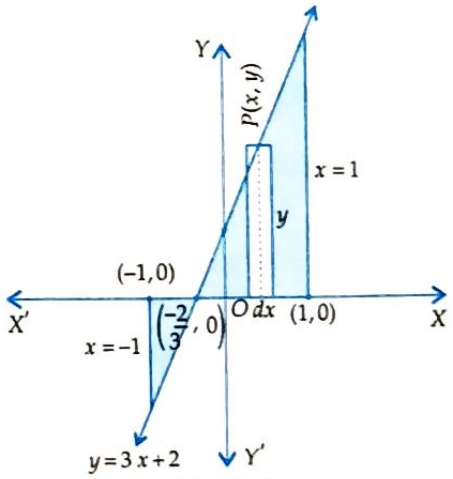


Fig. 20.12

$$\Rightarrow\ A = -\int_{-1}^{-2/3} (3x + 2)\, dx + \int_{-2/3}^{1} (3x + 2)\, dx$$

$$\Rightarrow\ A = -\left[\frac{3}{2}x^2 + 2x\right]_{-1}^{-2/3} + \left[\frac{3}{2}x^2 + 2x\right]_{-2/3}^{1} = \frac{1}{6} + \frac{25}{6} = \frac{13}{3} \text{ sq. units.}$$

EXAMPLE 9 *Using integration, find the area of the region bounded by the following curves, after making a rough sketch:* $y = 1 + |x+1|$, $x = -3, x = 3, y = 0$. [NCERT EXEMPLAR, CBSE 2014]

SOLUTION We have,

$$y = 1 + |x+1| \Rightarrow y = \begin{cases} 1 + x + 1 & , \text{ if } x + 1 \geq 0 \\ 1 - (x+1), & \text{if } x + 1 < 0 \end{cases} = \begin{cases} x + 2, & \text{if } x \geq -1 \\ -x & , \text{ if } x < -1 \end{cases}$$

Thus, the equations of the given curves are

$$y = x + 2 \text{ for } x \geq -1, \; y = -x \text{ for } x < -1, \; x = -3, x = 3 \text{ and } y = 0$$

Clearly, $y = x + 2$ is a straight line cutting x and y-axes at $(-2, 0)$ and $(0, 2)$ respectively. So, $y = x + 2$, for $x > -1$ represents that part of the line which is on the right side of $x = -1$.

The equation $y = -x$ for $x < -1$ represents that part of the line passing through the origin and making an angle of 135° with x-axis which is on the left side of $x = -1$. Clearly, $x = -3$ and $x = 3$ are lines parallel to y-axis. The region bounded by the given curves is shaded portion shown in Fig. 20.13.

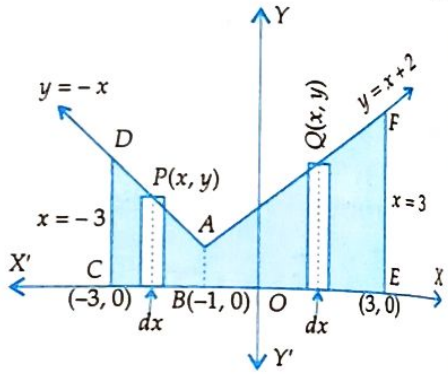


Fig. 20.13

When we slice the shaded region into vertical strips, we observe that vertical strips change their character at point A.

So, required area = Area $CDAB$ + Area $ABEF$.

Area CDAB: In area $CDAB$, each vertical strip has its upper end on $y = -x$ and lower end on x-axis. So, the approximating rectangle shown in Fig. 20.13 has, length $= |y|$, width $= dx$ and area $= |y|\,dx$. Since the approximating rectangle can move from $x = -3$ to $x = -1$.

$$\therefore \quad \text{Area } CDAB = \int_{-3}^{-1} |y|\,dx = \int_{-3}^{-1} y\,dx = \int_{-3}^{-1} -x\,dx \qquad [\because P(x, y) \text{ lies on } y = -x]$$

Let A denote the required area. Then,

Area ABEF: In area $ABEF$, each vertical strip has its lower end on x-axis and the upper end on $y = x + 2$. So, the approximating rectangle has,

length $= |y|$, width $= dx$ and area $= |y|\,dx$. As it can move from $x = -1$ to $x = 3$.

$$\therefore \quad \text{Area } ABEF = \int_{-1}^{3} |y|\,dx = \int_{-1}^{3} y\,dx = \int_{-1}^{3} (x+2)\,dx \qquad [\because Q(x, y) \text{ lies on } y = x + 2]$$

$$\therefore \quad A = \text{Area } CDAB + \text{Area } ABEF$$

$$\Rightarrow \quad A = \int_{-3}^{-1} -x\,dx + \int_{-1}^{3} (x+2)\,dx = -\left[\frac{x^2}{2}\right]_{-3}^{-1} + \left[\frac{x^2}{2} + 2x\right]_{-1}^{3}$$

$$\Rightarrow \quad A = -\left\{\frac{1}{2} - \frac{9}{2}\right\} + \left\{\left(\frac{9}{2} + 6\right) - \left(\frac{1}{2} - 2\right)\right\} = 16 \text{ sq. units}$$

EXAMPLE 10 *Sketch the graph* $y = |x+1|$. *Evaluate* $\int_{-3}^{1} |x+1|\,dx$. *What does this value represent on the graph?*

SOLUTION The equation of the given curve is

$$y = |x+1| = \begin{cases} x+1, & \text{if } x+1 \geq 0 \\ -(x+1), & \text{if } x+1 < 0 \end{cases} \quad \text{or,} \quad y = |x+1| = \begin{cases} x+1, & \text{if } x \geq -1 \\ -x-1, & \text{if } x < -1 \end{cases}$$

Clearly, $y = x+1$ is a straight line cutting x and y-axes at $(-1, 0)$ and $(0, 1)$ respectively. So, $y = x+1, x \geq -1$ represents that portion of the line which lies on the right side of $x = -1$. Similarly, $y = -x-1, x < -1$ represents that part of the line $y = -x-1$ which is on the left side of $x = -1$.

A rough sketch of $y = |x+1|$ is shown in Fig. 20.14. Clearly, $x = -3$ and $x = 1$ are lines parallel to y-axis. The region bounded by $y = |x+1|$, $x = -3$, $x = 1$ and x-axis is shaded in Fig. 20.14. The area of this region is given by $\int_{-3}^{1} |x+1|\, dx$.

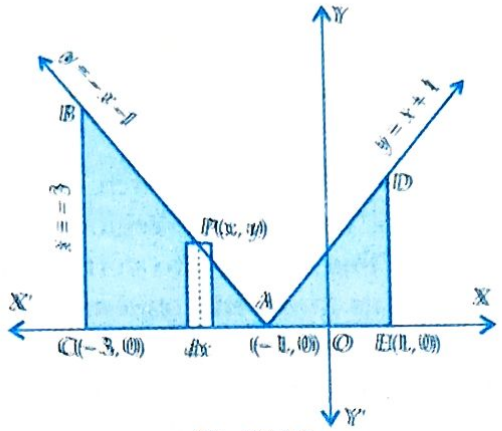


Fig. 20.14

Now, $\int_{-3}^{1} |x+1|\, dx = \int_{-3}^{-1} |x+1|\, dx + \int_{-1}^{1} |x+1|\, dx = \int_{-3}^{-1} -(x+1)\, dx + \int_{-1}^{1} (x+1)\, dx$

$$= -\left[\frac{(x+1)^2}{2}\right]_{-3}^{-1} + \left[\frac{(x+1)^2}{2}\right]_{-1}^{1} = -\left[0 - \frac{4}{2}\right] + \left[\frac{4}{2} - 0\right] = 4$$

This value represents the area of the shaded region shown in Fig. 20.14.

EXAMPLE 11 *Using integration find the area of the triangle formed by positive x-axis and tangent and normal to the circle* $x^2 + y^2 = 4$ *at* $(1, \sqrt{3})$. **[CBSE 2015]**

SOLUTION The equation of the circle is $x^2 + y^2 = 4$. Differentiating with respect to x, we obtain

$$2x + 2y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x}{y} \Rightarrow \left(\frac{dy}{dx}\right)_{(1, \sqrt{3})} = -\frac{1}{\sqrt{3}}$$

So, the equations of tangent and normal to the circle $x^2 + y^2 = 4$ at $(1, \sqrt{3})$ are

$$y - \sqrt{3} = -\frac{1}{\sqrt{3}}(x-1) \text{ and } y - \sqrt{3} = \sqrt{3}(x-1)$$

respectively or, $x + \sqrt{3}y = 4$ and $y = \sqrt{3}x$.

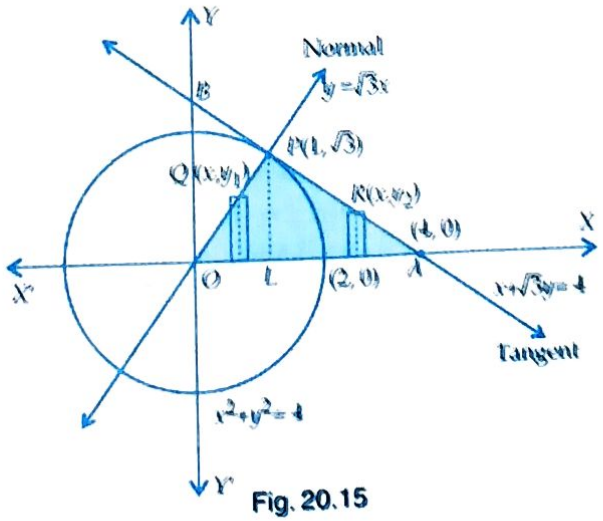


Fig. 20.15

The triangular region bounded by the positive x-axis, tangent and normal to the circle is the shaded region in Fig. 20.15. Let A be its area. Then,

$$A = \text{Area of the region } OPLO + \text{Area of the region } LPAL = \int_0^1 y_1\, dx + \int_1^4 y_2\, dx$$

Points $Q(x, y_1)$ and $R(x, y_2)$ lie on $y = \sqrt{3}x$ and $x + \sqrt{3}y = 4$ respectively. Therefore, $y_1 = \sqrt{3}$ and $x + \sqrt{3}\, y_2 = 4$.

$$\therefore \quad A = \int_0^1 \sqrt{3}\, x\, dx + \int_1^4 \frac{4-x}{\sqrt{3}}\, dx = \sqrt{3}\left[\frac{x^2}{2}\right]_0^1 + \frac{1}{\sqrt{3}}\left[4x - \frac{x^2}{2}\right]_1^4 = \left(\frac{\sqrt{3}}{2} + \frac{3\sqrt{3}}{2}\right) = 2\sqrt{3} \text{ sq. units}$$

BASED ON LOWER ORDER THINKING SKILLS (LOTS)

EXAMPLE 12 *Find the area bounded by the curve $y = \sin x$ between $x = 0$ and $x = 2\pi$.* [NCERT EXEMPLAR]

SOLUTION A rough sketch of the curve $y = \sin x$ between $x = 0$ and $x = 2\pi$ is shown in Fig. 20.16. The required region is shaded in Fig. 20.16. We slice this region into vertical strips. Each vertical strip has its lower end on x-axis and the upper end on $y = \sin x$. So, the approximating rectangle shown in Fig. 20.16 has, length $= |y|$. width $= dx$, and area $= |y|\, dx$. The approximating rectangle can move horizontally between $(0, 0)$ and $(2\pi, 0)$.

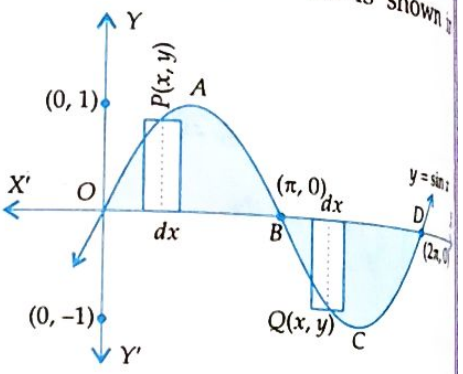


Fig. 20.16

So, required area A is given by

$$A = \int_0^{2\pi} |y|\, dx = \int_0^{\pi} |y|\, dx + \int_{\pi}^{2\pi} |y|\, dx$$

$$\Rightarrow \quad A = \int_0^{\pi} y\, dx + \int_{\pi}^{2\pi} -y\, dx \qquad \left[\because\ y \geq 0 \text{ for } 0 \leq x \leq \pi \text{ and } y < 0 \text{ for } \pi < x < 2\pi \right.$$
$$\left. \therefore\ |y| = y \text{ for } 0 \leq x \leq \pi \text{ and } |y| = -y \text{ for } \pi < x < 2\pi \right]$$

$$\Rightarrow \quad A = \int_0^{\pi} \sin x\, dx + \int_{\pi}^{2\pi} -\sin x\, dx \qquad [\because\ P(x, y) \text{ lies on } y = \sin x]$$

$$\Rightarrow \quad A = \left[-\cos x\right]_0^{\pi} + \left[\cos x\right]_{\pi}^{2\pi} = 2 + 2 = 4 \text{ sq. units}$$

EXAMPLE 13 *Find the area bounded by the curve $y = x|x|$, x-axis and the ordinates $x = -3$ and $x = 3$.*

SOLUTION The equation of the curve is $y = x|x| = \begin{cases} x^2, & x \geq 0 \\ -x^2, & x < 0 \end{cases}$ [CBSE 2024]

The graph of $y = x|x|$ is shown in Fig. 20.17 and the region bounded by $y = x|x|$, x-axis and the ordinates $x = -3$ and $x = 3$ is shaded in Fig. 20.17. Clearly, $y = x|x|$, being an odd function is symmetric in opposite quadrants. Therefore, required area is twice of the area of the shaded region in first quadrant.

Let us slice the region in first quadrant into vertical strips. The approximating rectangle shown in Fig. 20.17 has length $= |y_1|$, width $= dx$ and area $= |y_1|\, dx$. As it can move between $x = 0$ and $x = 3$. Therefore, area of the shaded region in first quadrant

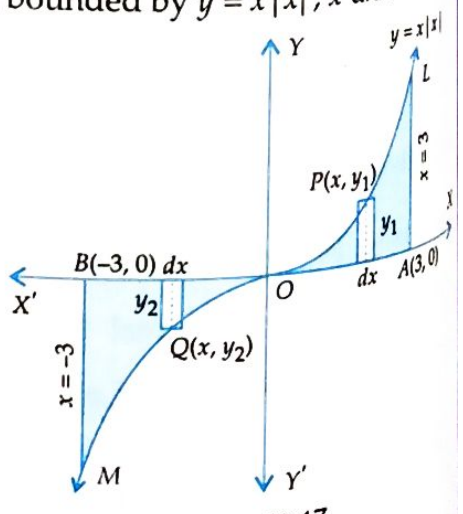


Fig. 20.17

$$= \int_0^3 |y_1|\, dx = \int_0^3 y_1\, dx \qquad [\because\ y_1 \geq 0 \ \therefore\ |y_1| = y_1]$$

$$\Rightarrow \quad A = \int_0^3 x^2\,dx = \left[\frac{x^3}{3}\right]_0^3 = 9 \text{ sq. units.} \qquad \left[\because P(x, y_1) \text{ lies on } y = x^2 \therefore\ y_1 = x^2\right]$$

Hence, required area $= 2A = 2 \times 9 = 18$ sq. units.

ALTER The graph of $y = x|x|$ is shown in Fig. 20.17. We have to find the area of the shaded region. Clearly,

Required area = Area of region $OBMO$ + Area of region $OALO$.

Area of region OBMO: Let us slice the region $OBMO$ into vertical strips. The approximating rectangle shown in Fig. 20.17 has length $= |y_2|$, width $= dx$ and area $= |y_2|\,dx$. Clearly, it can move horizontally between $x = -3$ and $x = 0$.

$$\therefore \quad \text{Area of region } OBMO = \int_{-3}^0 |y_2|\,dx = \int_{-3}^0 -y_2\,dx \qquad \left[\because y_2 < 0 \ \therefore\ |y_2| = -y_2\right]$$

$$= \int_{-3}^0 -(-x^2)\,dx \qquad \left[\because Q(x, y_2) \text{ lies on } y = -x^2 \ \therefore\ y_2 = -x^2\right]$$

$$= \int_{-3}^0 x^2\,dx = \left[\frac{x^3}{3}\right]_{-3}^0 = 0 - (-9) = 9 \text{ sq. units.}$$

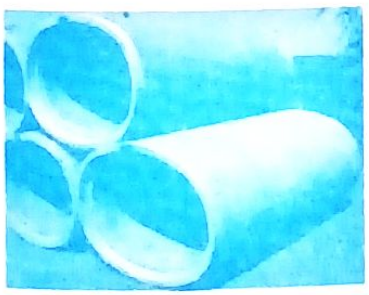
Area of region OALO: Let us slice the region $OALO$ into the vertical strips. The approximating rectangle shown in Fig. 20.17 has length $= |y_1|$, width $= dx$ and area $= |y_1|\,dx$. Clearly, it can move between $x = 0$ and $x = 3$.

$$\therefore \text{ Area of region } OALO = \int_0^3 |y_1|\,dx = \int_0^3 y_1\,dxS \qquad \left[\because y_1 > 0 \ \therefore\ |y_1| = y_1\right]$$

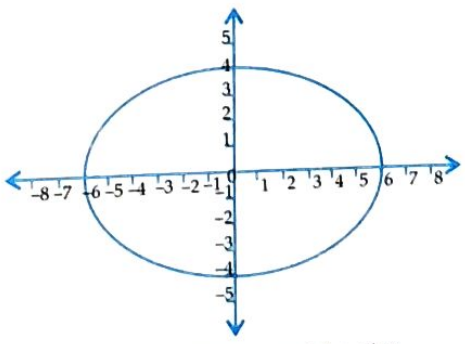
$$= \int_0^3 x^2\,dx = \left[\frac{x^3}{3}\right]_0^3 = 9 \text{ sq. units} \qquad \left[\because P(x, y_1) \text{ lies on } y = x^2 \therefore y_1 = x^2\right]$$

Hence, Required Area = Area of region $OBMO$ + Area of region $OALO = (9 + 9) = 18$ sq. units.

EXAMPLE 14 *Show below are concrete elliptical water pipes, each 10 feet in length.*



Concrete elliptical pipes



Inner circumference of the pipe

Fig. 20.18

The graph given above represents the inner circumference of the elliptical pipe, where x and y are in feet. Assume that the water flows uniformly and fully covers the inner cross-sectional area of the pipe. Find the volume of water in the pipe at a given instant of time, in terms of π by using the integration method.

SOLUTION Let the equation of the ellipse be $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. It is evident from the graph of the ellipse that $2a=12$ and $2b=8$ i.e. $a=6$ and $b=4$. So, the equation of the ellipse is $\frac{x^2}{36}+\frac{y^2}{16}=1$. Let A denote the area of the ellipse. Then, the volume V of the water in 10 ft long elliptical pipe is given by

$$V=10A$$

Now, $\frac{x^2}{36}+\frac{y^2}{16}=1 \Rightarrow \frac{y^2}{16}=1-\frac{x^2}{36} \Rightarrow y=\frac{2}{3}\sqrt{36-x^2}$ ...(i)

Using integration, the area A of the ellipse is given by

$$A=4\int_0^6 y\,dx=4\int_0^6 \frac{2}{3}\sqrt{36-x^2}\,dx=\frac{8}{3}\int_0^6 \sqrt{36-x^2}\,dx$$

$$\Rightarrow \quad A=\frac{8}{3}\left[\frac{1}{2}x\sqrt{36-x^2}+\frac{36}{2}\sin^{-1}\frac{x}{6}\right]_0^6=\frac{8}{3}(18\sin^{-1}1)=48\frac{\pi}{2}=24\pi \quad \text{...(ii)}$$

From (i) and (ii), we obtain: $V=10\times 24\pi=240\,\pi$ cubic feet.

BASED ON HIGHER ORDER THINKING SKILLS (HOTS)

EXAMPLE 15 *Using the method of integration find the area-bounded by the curve* $|x|+|y|=1$. [NCERT]

SOLUTION The equation of the curve is

$$|x|+|y| = 1 \Leftrightarrow \begin{cases} x+y=1 & \text{when } x\geq 0, y\geq 0 \\ -x+y=1 & \text{when } x<0, y\geq 0 \\ x-y=1 & \text{when } x\geq 0, y<0 \\ -x-y=1 & \text{when } x<0, y<0 \end{cases}$$

Lines $x+y=1, -x+y=1, x-y=1$ and $-x-y=1$ enclose a square shown in Fig. 20.19. The area A bounded by the square is given by

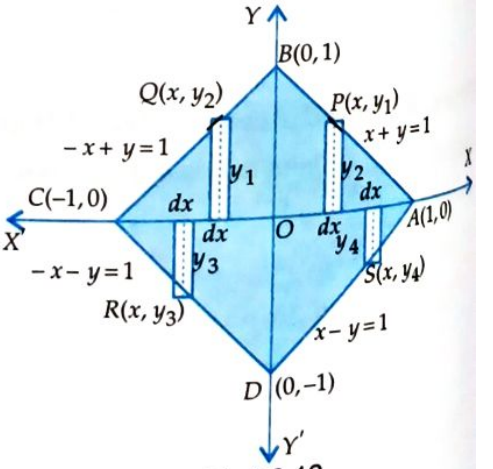


Fig. 20.19

$$A = \text{Area } COBO + \text{Area } OABO + \text{Area } CODC + \text{Area } ODAO$$

$$\Rightarrow A = \int_{-1}^{0}|y_1|\,dx+\int_0^1|y_2|\,dx+\int_{-1}^{0}|y_3|\,dx+\int_0^1|y_4|\,dx$$

$$\Rightarrow A = \int_{-1}^{0}|1+x|\,dx+\int_0^1|1-x|\,dx+\int_{-1}^{0}|-x-1|\,dx+\int_0^1|x-1|\,dx$$

$$\Rightarrow A = \int_{-1}^{0} (1+x)\,dx + \int_{0}^{1} (1-x)\,dx + \int_{-1}^{0} (x+1)\,dx + \int_{0}^{1} (1-x)\,dx$$

$$\Rightarrow A = 2\int_{-1}^{0} (x+1)\,dx + 2\int_{0}^{1} (1-x)\,dx = 2\left[\frac{x^2}{2}+x\right]_{-1}^{0} + 2\left[x-\frac{x^2}{2}\right]_{0}^{1}$$

$$\Rightarrow A = 2\left\{0-\left(\frac{1}{2}-1\right)\right\} + 2\left\{\left(1-\frac{1}{2}\right)-0\right\} = 1+1 = 2 \text{ sq. units.}$$

EXERCISE 20.1

BASIC

1. Using integration, find the area of the region bounded between the line $x = 2$ and the parabola $y^2 = 8x$.
2. Using integration, find the area of the region bounded by the line $y - 1 = x$, the x –axis and the ordinates $x = -2$ and $x = 3$. [CBSE 2002, 2022]
3. If the area of the region bounded by the curve $y^2 = 4ax$ and the line $x = 4a$ is $\frac{256}{3}$ sq. units, then using integration, find the value of a, where $a > 0$. [CBSE 2022]
4. Find the area lying above the x-axis and under the parabola $y = 4x - x^2$.
5. Draw a rough sketch to indicate the region bounded between the curve $y^2 = 4x$ and the lines $x = 0$ and $x = 3$. Also, find the area of this region. [CBSE 2022]
6. Make a rough sketch of the graph of the function $y = 4 - x^2$, $0 \le x \le 2$ and determine the area enclosed by the curve, the x-axis and the lines $x = 0$ and $x = 2$.
7. Sketch the graph of $y = \sqrt{x+1}$ in $[0, 4]$ and determine the area of the region enclosed by the curve, the x-axis and the lines $x = 0$, $x = 4$.
8. Find the area under the curve $y = \sqrt{6x+4}$ above x-axis from $x = 0$ to $x = 2$. Draw a sketch of curve also.
9. Using integration, find the area of the ellipse $\frac{x^2}{16} + \frac{y^2}{4} = 1$, included between the lines $x = -2$ and $x = 2$. [CBSE 2024]
10. Draw a rough sketch of the graph of the curve $\frac{x^2}{4} + \frac{y^2}{9} = 1$ and evaluate the area of the region under the curve and above the x-axis. [NCERT]
11. Sketch the region $\{(x, y) : 4x^2 + y^2 = 36\}$ and find the area of the region enclosed by it, using integration. [CBSE 2024]
12. Using integration, find the area of the region enclosed by the line $y = \sqrt{3}x$, semi-circle $y = \sqrt{4 - x^2}$ and x-axis in first quadrant. [CBSE 2022]
13. Determine the area under the curve $y = \sqrt{a^2 - x^2}$ included between the lines $x = 0$ and $x = a$. [NCERT EXEMPLAR]
14. Using integration, find the area of the region bounded by the line $y = mx$ $(m > 0)$, x-axis and the lines $x = 1$ and $x = 2$. [CBSE 2023]
15. Using definite integrals, find the area of the circle $x^2 + y^2 = a^2$.

28. Required area $= 2\int_{a/2}^{a} \sqrt{a^2 - x^2}\, dx$

$$= \left[x\sqrt{a^2 - x^2} + a^2 \sin^{-1}\frac{x}{a} \right]_{a/2}^{a} = \frac{a^2}{2}(4\pi - 3\sqrt{3})$$

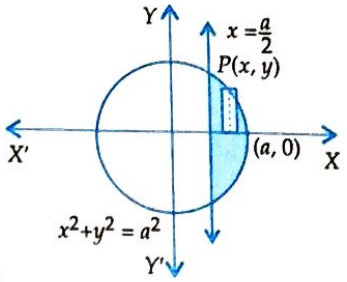


Fig. 20.25

29. Required area $= 2\int_{a}^{4a} y\, dx = 2\int_{a}^{4a} \sqrt{4ax}\, dx = 4\sqrt{a}\left[\frac{2}{3}x^{3/2}\right]_{a}^{4a} = \frac{56}{3}a^2$

OR

Required area $= 2\int_{1}^{2} y\frac{dx}{dt}\, dt = 2\int_{1}^{2} (2at)(2at)\, dt = 8a^2\int_{1}^{2} t^2\, dt$ $[\because x = at^2, y = 2at \Rightarrow y^2 = 4ax]$

20.4 AREA USING HORIZONTAL STRIPS

In order to find the areas of some regions, it is easier to form horizontal strips rather than vertical strips discussed in earlier section. The procedure for horizontal strips is analogous to the procedure for vertical strips.

In order to find the area of the region bounded by the curve $x = f(y)$, y-axis and the lines $y = c$ and $y = d$, we may use the following algorithm.

ALGORITHM

Step I *Make sketch of the curve and identify the region whose area is to be found.*

Step II *Slice the region into horizontal strips. Take an arbitrary point $P(x,y)$ on the curve and construct a representative strip of width dy and $(y, 0)$ as the mid-point of its base.*

Step III *Construct an approximating rectangle whose base is same as that of the representative strip and length equal to $|x| = |f(y)|$.*

Step IV *Find the area of the approximating rectangle as $|x|\, dy = |f(y)|\, dy$.*

Step V *Find the values of y, say $y = c$ and $y = d$ within which the approximating rectangle can move vertically in the given region and form the integral* $\int_{c}^{d} |x|\, dy = \int_{c}^{d} |f(y)|\, dy$.

Step VI *Evaluate the integral obtained in step V. The value so obtained is the required area.*

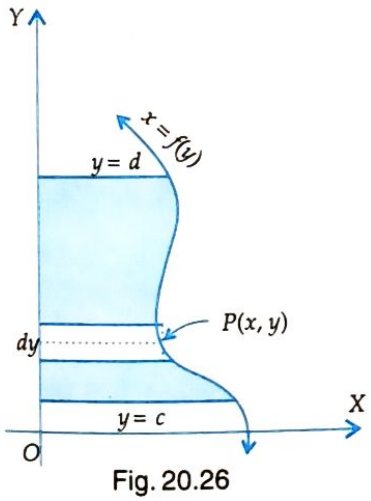


Fig. 20.26

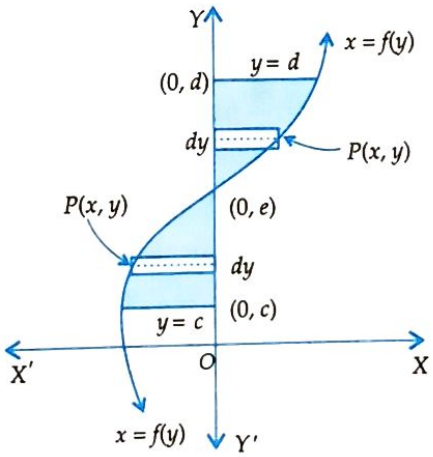


Fig. 20.27

REMARK 1 *If the function $f(y)$ is defined on $[c, d]$ on y-axis such that the curve $x = f(y)$ lies on the left of y-axis on $[c, e]$ and on the right of y-axis on $[e, d]$ as shown in Fig. 20.27, then area of the region bounded by the curve $x = f(y)$, y-axis and lines $y = c$ and $y = d$ is given by*

$$A = \int_c^d |x|\,dy = \int_c^e |x|\,dy + \int_e^d |x|\,dy$$

or, $$A = \int_c^d |f(y)|\,dy = \int_c^e |f(y)|\,dy + \int_e^d |f(y)|\,dy$$

or, $$A = \int_c^e -f(y)\,dy + \int_e^d f(y)\,dy \quad [\because\ x = f(y) < 0 \text{ for } c < y < e \text{ and } x = f(y) > 0 \text{ for } e < y < d]$$

ILLUSTRATIVE EXAMPLES

BASED ON BASIC CONCEPTS (BASIC)

EXAMPLE 1 *Find the area of the region bounded by the curve* $y^2 = 2y - x$ *and the y-axis.*

SOLUTION The equation of the given curve is

$$y^2 = 2y - x \Rightarrow y^2 - 2y = -x \Rightarrow y^2 - 2y + 1 = -x + 1 \Rightarrow (y-1)^2 = -(x-1)$$

Clearly, this equation represents a parabola with vertex at (1, 1) and opens on the left. Putting $x = 0$ in $y^2 = 2y - x$, we get: $y^2 - 2y = 0 \Rightarrow y = 0, 2$.

So, the curve meets y-axis at (0, 0) and (0, 2). A rough sketch of the curve is as shown in Fig. 20.28 and the area of the region bounded by the curve and y-axis is shaded in Fig. 20.28. Here, we slice the shaded region into horizontal strips. For the approximating rectangle shown in Fig. 20.28, we have width $= dy$, length $= |x|$, and area $= |x|\,dy$. The approximating rectangle can move vertically from $y = 0$ to $y = 2$. So, required area A is given by

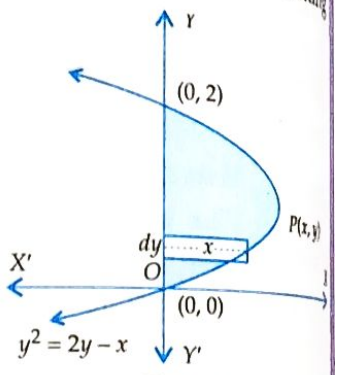


Fig. 20.28

$$\therefore \quad A = \int_0^2 |x|\,dy = \int_0^2 x\,dy \qquad [\because x \geq 0 \ \therefore |x| = x]$$

$$\Rightarrow \quad A = \int_0^2 (2y - y^2)\,dy = \left[y^2 - \frac{y^3}{3}\right]_0^2 \qquad [\because P(x, y) \text{ lies on } y^2 = 2y - x \therefore x = 2y - y^2]$$

$$\Rightarrow \quad A = 4 - \frac{8}{3} = \frac{4}{3} \text{ sq. units.}$$

EXAMPLE 2 *Sketch the region lying in the first quadrant and bounded by* $y = 9x^2$, $x = 0$, $y = 1$ *and* $y = 4$. *Find the area of the region using integration.*

SOLUTION The equation $y = 9x^2$ represents an upward opening parabola with axis as y-axis and vertex at the origin. Clearly, the shaded region is the region lying in the first quadrant and bounded by $y = 9x^2$, $x = 0$, $y = 1$ and $y = 4$. Let us slice this region into horizontal strips. The approximating rectangle shown in Fig. 20.29 has length $= |x|$ width $= dy$ and area $= |x|\,dy$. Clearly, it can move vertically between $y = 1$ and $y = 4$. So, required area A is given by

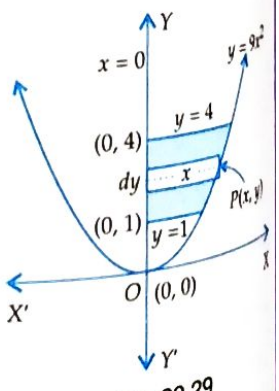


Fig. 20.29

$$A = \int_1^4 |x|\, dy = \int_1^4 x\, dy \qquad [\because\ x \geq 0\ \therefore\ |x| = x]$$

$$\Rightarrow A = \int_1^4 \sqrt{\frac{y}{9}}\, dy = \frac{1}{3}\int_1^4 \sqrt{y}\, dy = \frac{1}{3}\times\frac{2}{3}\left[y^{3/2}\right]_1^4 \qquad \left[\because P(x, y) \text{ lies on } y = 9x^2\ \therefore\ x = \sqrt{\frac{y}{9}}\right]$$

$$\Rightarrow A = \frac{2}{9}(8-1) = \frac{14}{9} \text{ sq. units}$$

EXAMPLE 3 *Find the area bounded by the curve $y^2 = 4ax$ and the lines $y = 2a$ and y-axis.*

SOLUTION Clearly, the equation $y^2 = 4ax$ represents a parabola with vertex $(0, 0)$ and axis as x-axis. The equation $y = 2a$ represents a straight line parallel to x-axis at a distance $2a$ from it as shown in Fig. 20.30. The required region is the shaded portion in Fig. 20.30. To find the area of the shaded region shown in Fig. 20.30, we slice it into horizontal strips. We observe that each horizontal strip has its left end on y-axis and the right end on the parabola $y^2 = 4ax$. So, the approximating rectangle shown in Fig. 20.30 has its length $= |x|$, width $= dy$ and area $= |x|\, dy$.

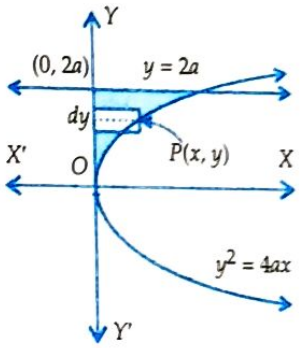


Fig. 20.30

Since the approximating rectangle can move vertically from $y = 0$ to $y = 2a$. So, required area A is given by

$$A = \int_0^{2a} |x|\, dy = \int_0^{2a} x\, dy \qquad [\because\ x \geq 0\ \therefore\ |x| = x]$$

$$\Rightarrow \quad A = \int_0^{2a} \frac{y^2}{4a}\, dy = \frac{1}{4a}\left[\frac{y^3}{3}\right]_0^{2a} \qquad \left[\because P(x, y) \text{ lies on } y^2 = 4ax\ \therefore\ x = \frac{y^2}{4a}\right]$$

$$\Rightarrow \quad A = \frac{1}{4a}\left(\frac{8a^3}{3} - 0\right) = \frac{2a^2}{3} \text{ sq. units}$$

BASED ON LOWER ORDER THINKING SKILLS (LOTS)

EXAMPLE 4 *Find the area of the region bounded by $y = -1, y = 2, x = y^3$ and $x = 0$.*

SOLUTION A rough sketch of the curve $x = y^3$ as shown in Fig. 20.31. Clearly, $y = -1$ and $y = 2$ are straight lines parallel to x-axis. The required region is shaded in Fig. 20.31. When we slice this region into horizontal strips, we observe that each strip has its one end on y-axis and other end on $x = y^3$. So, the approximating rectangle shown in Fig. 20.31 has, length $= |x|$, width $= dy$ and area $= |x|\, dy$. Since the approximating rectangle can move vertically from $y = -1$ to $y = 2$. So, required area A is given by

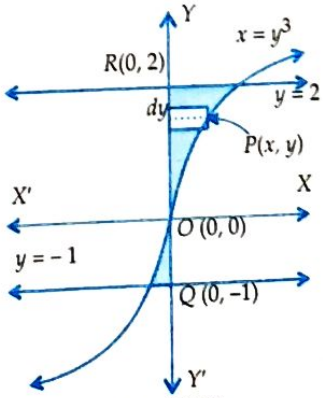


Fig. 20.31

$$A = \int_{-1}^{2} |x|\, dy = \int_{-1}^{0} |x|\, dy + \int_0^2 |x|\, dy$$

$$\Rightarrow \quad A = \int_{-1}^{0} -x\,dy + \int_{0}^{2} x\,dy \qquad [\because x < 0 \text{ for } -1 \le y < 0 \text{ and } x > 0 \text{ for } 0 \le y <$$

$$\Rightarrow \quad A = \int_{-1}^{0} -y^3\,dy + \int_{0}^{2} y^3 dy = -\left[\frac{y^4}{4}\right]_{-1}^{0} + \left[\frac{y^4}{4}\right]_{0}^{2} \qquad [\because P(x, y) \text{ lies on } x = y^3 \therefore x =$$

$$\Rightarrow \quad A = \frac{1}{4} + 4 = \frac{17}{4} \text{ sq. units}$$

EXERCISE 20.

BASIC

1. Find the area of the region in the first quadrant bounded by the parabola $y = 4x^2$ and the lines $x = 0$, $y = 1$ and $y = 4$.
2. Find the area of the region bounded by $x^2 = 16y$, $y = 1$, $y = 4$ and the y-axis in the first quadrant.
3. Find the area of the region bounded by $x^2 = 4ay$ and its latusrectum.
4. Find the area of the region bounded by $x^2 + 16y = 0$ and its latusrectum.

BASIC ON LOTS

5. Find the area of the region bounded by the curve $ay^2 = x^3$, the y-axis and the lines $y = a$ and $y = 2a$.
6. Sketch a graph of $y = x^2$. Using integration, find the area of the region bounded by $y = 1$, $x = 0$ and $y = x^2$. [CBSE 2015]

ANSWERS

1. $\frac{7}{3}$ sq. units
2. $\frac{56}{3}$ sq. units
3. $\frac{8a^2}{3}$ sq. units
4. $\frac{128}{3}$ sq. units
5. $\frac{3}{5}(2^{5/3} - 1)\,a^2$ sq. units
6. 18 sq. units

HINTS TO SELECTED PROBLEMS

5. Required area $= \int_{a}^{2a} x\,dy$

$$= \int_{a}^{2a} (ay^2)^{1/3}\,dy = a^{1/3} \int_{a}^{2a} y^{2/3}\,dy$$

$$= a^{1/3}\left[\frac{3}{5}y^{5/3}\right]_{a}^{2a}$$

$$= \frac{3}{5}a^{1/3}\left\{(2a)^{5/3} - a^{5/3}\right\} \text{ sq. units}$$

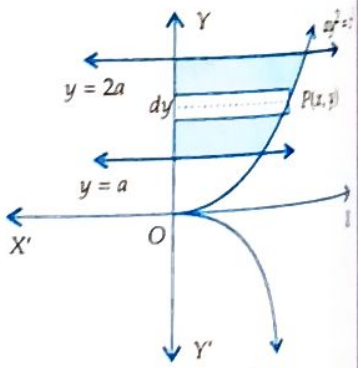


Fig. 20.32

20.5 AREA BETWEEN TWO CURVES BY USING VERTICAL STRIPS

Uptill now we have been discussing problems either on finding the areas of the regions bounded by a curve $y = f(x)$, x-axis and on the left and right by the vertical lines $x = a$ and $x =$ respectively or on finding the areas of the regions bounded by a curve $x = f(y)$, y-axis and between the horizontal lines $y = c$ and $y = d$ on bottom and top respectively. In some practical problems, we may be interested in computing the area between two curves one below the other or one on the left of other. If two curves are such that one is below the other and we wish to find the area of the region bounded by them and on the left and right by vertical lines. In such cases we may use the following algorithm.